

CU 211 and CU 212

Installation and operating instructions

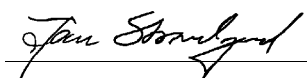


Declaration of Conformity

We **Grundfos** declare under our sole responsibility that the products **CU 211** and **CU 212**, to which this declaration relates, are in conformity with these Council directives on the approximation of the laws of the EC member states:

- **Low Voltage Directive** (2006/95/EC)
Standard used: EN 60335-1: 2002.
- **EMC Directive** (2004/108/EC)
Standards used: EN 61000-6-2: 2005 and EN 61000-6-3: 2007.

Bjerringbro, 15th September 2008



Jan Strandgaard
Technical Director

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document

**Warning**

If these safety instructions are not observed, it may result in personal injury!



If these safety instructions are not observed, it may result in malfunction or damage to the equipment!



Notes or instructions that make the job easier and ensure safe operation.

**Warning**

These instructions must be observed for explosion-proof pumps. It is advisable also to follow these instructions for standard pumps.

2. General description

The control units CU 211 and CU 212 are designed for control of pumps in wastewater, water supply and draining systems.

CU 211 is designed for control of one pump and CU 212 of two pumps.

CU 211 and CU 212 have an alarm output for common alarm and an alarm output for separate high-level alarm.

Furthermore, the control units incorporate a buzzer.

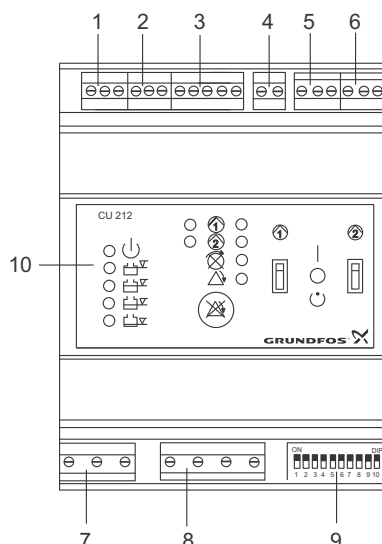
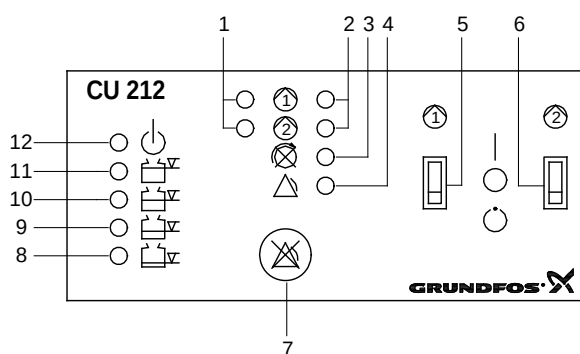


Fig. 1 CU 212

Pos.	Description
1	Input terminals for AC and DC level switches.
2	Input terminals for DC level switches.
3	Input terminals for alarm from pump sensors and motor protector.
4	Terminals for accumulator.
5	Output terminals for high-level alarm with NO/NC contact. See section 8. Overview of terminals.
6	Output terminals for common alarm with NO/NC contact. See section 8. Overview of terminals.
7	Input terminals for supply voltage.
8	Output terminals for pump 1 and pump 2. For CU 211 there is only an output for one pump.
9	DIP switch. See section 2.2 DIP switch.
10	Control panel. See section 2.1 Control panel.

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2.1 Control panel



TM01 6397 3902

Fig. 2 Control panel CU 212

Key to the symbols in fig. 2:

Pos.	Description
1	Green indicator lights for pumps 1 and 2; flashing for start delay, shining for pump operation. CU 211 has only one indicator light.
2	Red indicator lights for pumps 1 and 2; flashing for thermo fault, shining for fault in motor protector. CU 211 has only one indicator light.
3	Red indicator light; indicating wrong phase sequence (three-phase pumps only).
4	Red indicator light; shining for common alarm.
5	ON-OFF-AUTO selector switch for pump 1.
6	ON-OFF-AUTO selector switch for pump 2. CU 211 has no selector switch for pump 2.
7	Reset button. Push-button for manual resetting of external alarm devices and the built-in buzzer. See section 2.1.1 <i>Resetting of alarm</i> .
8	Orange indicator light; shining for liquid level 1. See section 4. <i>Application</i> .
9	Orange indicator light; shining for liquid level 2. See section 4. <i>Application</i> .
10	Orange indicator light; shining for liquid level 3. See section 4. <i>Application</i> .
11	Orange indicator light; shining for liquid level 4, flashing for high-level alarm. See section 4. <i>Application</i> .
12	Green indicator light; shining when the power supply has been switched on.

2.1.1 Resetting of alarm



The **reset button** is a push-button for manual resetting of external alarm devices and the built-in buzzer. The button is NOT for resetting of alarm indications; do this by means of the ON-OFF-AUTO selector switch. Even if the fault condition still exists, the external alarm devices and the built-in buzzer will be reset when the reset button is pressed.

2.2 DIP switch

CU 211 and CU 212 have a 10-pole DIP switch positioned in the bottom right corner. See fig. 3.

Caution

When setting the DIP switch, the control unit must have been switched off for at least 1 minute.

Each individual switch (1 to 10) of the DIP switch can be set to position OFF or ON.

Note

The DIP switch must not be set to other switch combinations than those described in section 5. Functions.

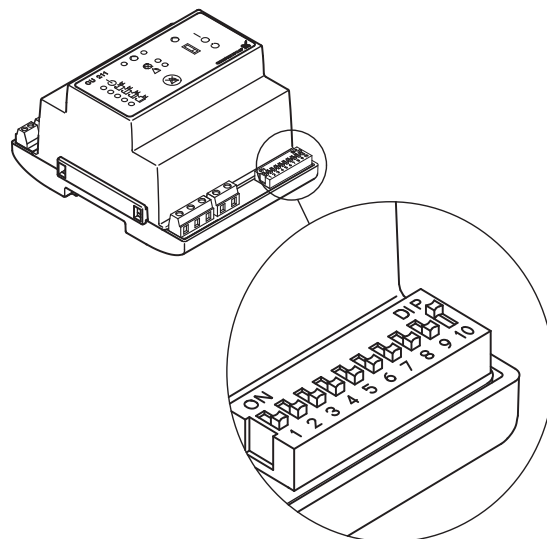


Fig. 3 DIP switch setting

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2.3 Dimensions

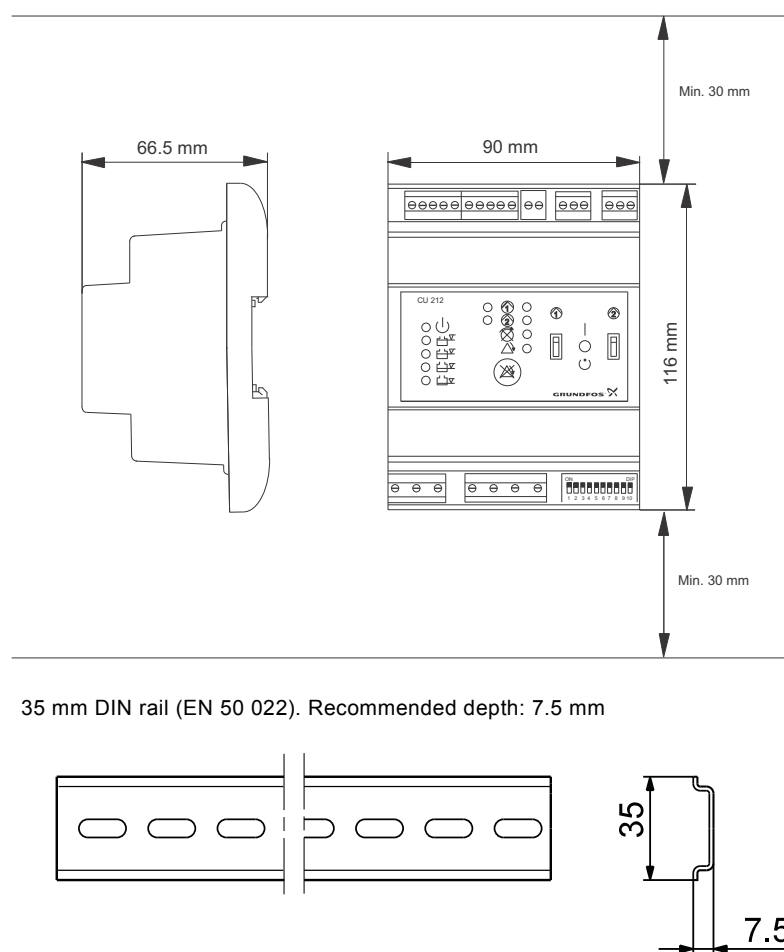


Fig. 4 Dimensional sketch of control unit and DIN rail

3. Identification

3.1 Nameplate

The nameplate is fitted on the side of the control unit.

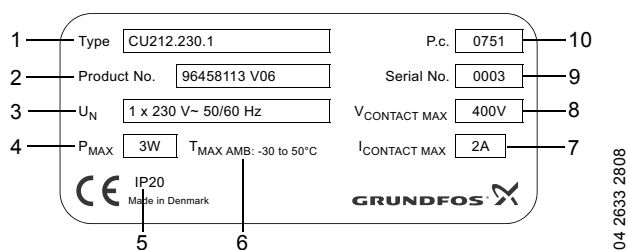


Fig. 5 Nameplate, CU 212

Key to the nameplate:

Pos.	Description
1	Type designation
2	Product number
3	Rated voltage and frequency
4	Max. power
5	Enclosure class
6	Max. ambient temperature
7	Contact current
8	Contact voltage
9	Serial number
10	Production code (year-week code)

3.2 Type key

Example	CU	212	.230	.1
Grundfos control unit				
Variant designation				
211				
212 ¹⁾				
Voltage [V]				
Phases:				
1 = single-phase				
3 = three-phase				

- ¹⁾ 211 = single-pump control
212 = two-pump control.

4. Application

CU 211 and 212 can be used for level control in a wide range of industrial, private and public applications.

Examples:

- tanks in industry
- wastewater collection tanks
- lifting stations (with more than one pump).

CU 211 can be set to operation/control in seven different applications.

CU 212 can be set to operation/control in four different applications.

See section 5. *Functions* for additional information.

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5. Functions

This section describes a number of systems for which CU 211 and CU 212 are designed and the associated DIP switch settings.

5.1 Two float switches and parallel operation with three float switches

Use **CU 211** in systems with one pump and two float switches. The pump is controlled by the liquid level in the tank.

- When the float switch, pos. 1, is lifted up by the liquid, the pump is started.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the settable stop delay sets in, and after that the pump is stopped.
- When the float switch, pos. 2, is lifted up by the liquid, the high-level alarm is activated.

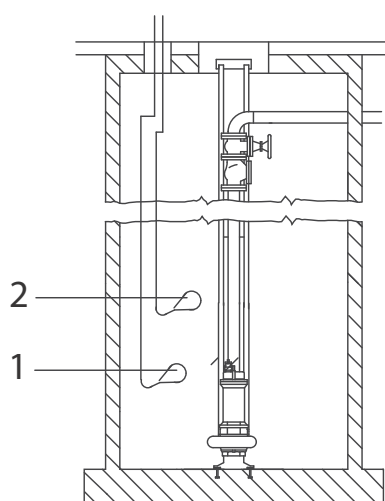


Fig. 6 Systems with one pump and two float switches

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Use **CU 212** in systems with two pumps in parallel operation and three float switches. The pumps are controlled by the liquid level in the tank.

- When the float switch, pos. 1, is lifted up by the liquid, the first pump is started.
- When the float switch, pos. 2, is lifted up by the liquid, the second pump is started.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the settable stop delay sets in, and after that both pumps are stopped.
- The pumps alternate automatically.
- When the float switch, pos. 3, is lifted up by the liquid, the high-level alarm is activated.

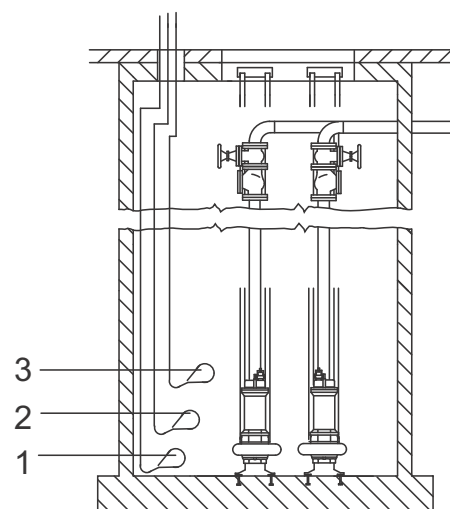


Fig. 7 Systems with two pumps in parallel operation and three float switches

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5.1.1 DIP switch setting

The DIP switch settings apply to both system types and to both CU 211 and CU 212.

- Switches **1, 2 and 3**: System type



This setting encodes the system type.
CU 211: Two float switches.
CU 212: Parallel operation with three float switches.

- Switch **4**: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

- Switches **5, 6 and 7**: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.
The pump must not run dry.

0 sec.



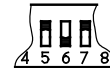
60 sec.



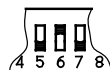
15 sec.



90 sec.



30 sec.



120 sec.



45 sec.



180 sec.



- Switch **8**:



Switch 8 has no function in connection with these system types, but keep the setting in this position.

- Switch **9**: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch **10**: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has cut out the pump. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.2 Three float switches and parallel operation with four float switches

Use **CU 211** in systems with one pump and three float switches. The pump is controlled by the liquid level in the tank.

- When the float switch, pos. 2, is lifted up by the liquid, the pump is started.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the pump is stopped. It is possible to set a stop delay which delays the stop of the pump.
- When the float switch, pos. 3, is lifted up by the liquid, the high-level alarm is activated.

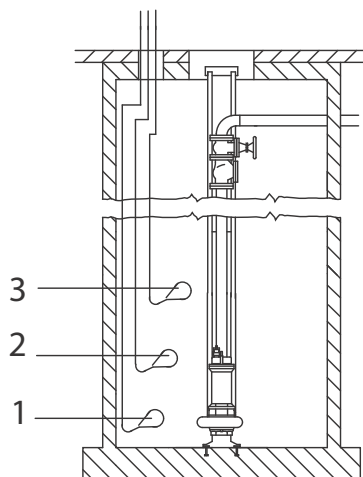


Fig. 8 Systems with one pump and three float switches

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Use **CU 212** in systems with two pumps in parallel operation and four float switches. The pumps are controlled by the liquid level in the tank.

- When the float switch, pos. 2, is lifted up by the liquid, the first pump is started.
- When the float switch, pos. 3, is lifted up by the liquid, the second pump is started.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the settable stop delay sets in, and after that both pumps are stopped.
- The pumps alternate automatically.
- When the float switch, pos. 4, is lifted up by the liquid, the high-level alarm is activated.

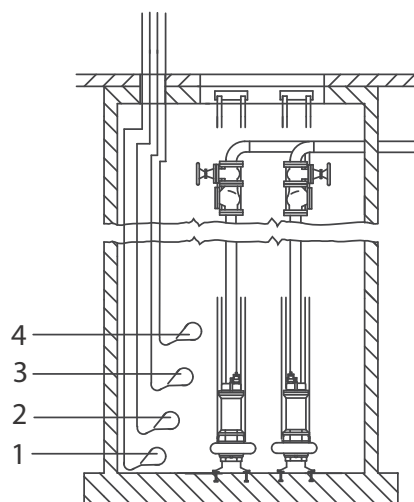


Fig. 9 Systems with two pumps in parallel operation and four float switches

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5.2.1 DIP switch setting

The DIP switch settings apply to both system types and to both CU 211 and CU 212.

- Switches **1, 2 and 3**: System type



This setting encodes the system type.
CU 211: Three float switches.
CU 212: Parallel operation with four float switches.

- Switch **4**: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

- Switches **5, 6 and 7**: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.
The pump must not run dry.

0 sec.



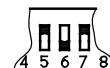
60 sec.



15 sec.



90 sec.



30 sec.



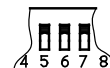
120 sec.



45 sec.



180 sec.



- Switch **8**:



Switch 8 has no function in connection with these system types, but keep the setting in this position.

- Switch **9**: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch **10**: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.3 Four float switches and parallel operation with 100 % standby operation

Use **CU 211** in systems with one pump and four float switches. The pump is controlled by the liquid level in the tank.

- When the float switch, pos. 3, is lifted up by the liquid, the pump is started.
- When the float switch, pos. 2, is no longer lifted up by the liquid, the pump is stopped. It is possible to set a stop delay which delays the stop of the pump.
- When the float switch, pos. 4, is lifted up by the liquid, the high-level alarm is activated.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the dry running alarm is activated.

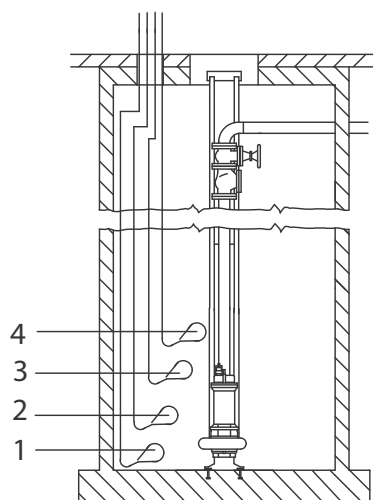


Fig. 10 Systems with one pump and four float switches

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Use **CU 212** in systems with two pumps in parallel operation and four float switches with 100 % standby operation. The pumps are controlled by the liquid level in the tank.

- When the float switch, pos. 2, is lifted up by the liquid, the first pump is started.
- When the float switch, pos. 3, is lifted up by the liquid, the second pump is started.
- When the float switch, pos. 1, is no longer lifted up by the liquid, the settable stop delay sets in, and after that both pumps are stopped.
- The pumps alternate automatically.
- When the float switch, pos. 4, is lifted up by the liquid, the high-level alarm is activated.

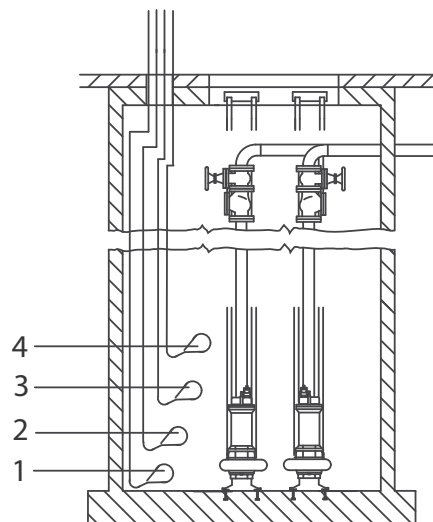


Fig. 11 Systems with two pumps in parallel operation with 100 % standby operation and four float switches

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5.3.1 DIP switch setting

The DIP switch settings apply to both system types and to both CU 211 and CU 212.

- Switches 1, 2 and 3: System type



This setting encodes the system type.
CU 211: Four float switches.
CU 212: 100 % standby operation.

- Switch 4: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

- Switches 5, 6 and 7: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.

The pump must not run dry.

0 sec.



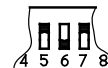
60 sec.



15 sec.



90 sec.



30 sec.



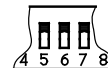
120 sec.



45 sec.



180 sec.



- Switch 8:



Switch 8 has no function in connection with these system types, but keep the setting in this position.

- Switch 9: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch 10: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



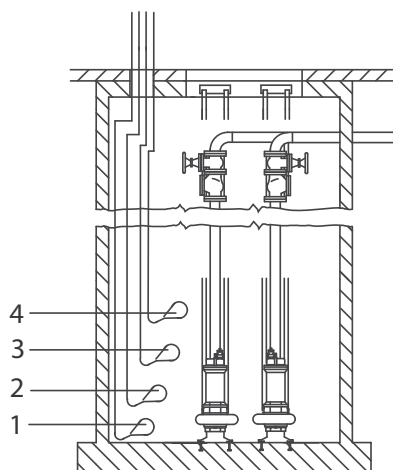
Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.4 Four float switches and parallel operation

Use **CU 212** in systems with two pumps and four float switches in so-called "full-control operation". The pumps are controlled by the liquid level in the tank.

- When the float switch, pos. 3, is lifted up by the liquid, the first pump is started.
 - When the float switch, pos. 4, is lifted up by the liquid, the second pump is started.
 - When the float switch, pos. 2, is no longer lifted up by the liquid, the second pump is stopped.
 - When the float switch, pos. 1, is no longer lifted up by the liquid, the first pump is stopped.
- It is possible to set a stop delay which delays the stop of the pumps.
- The pumps alternate automatically.



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Fig. 12 Systems with two pumps and four float switches in parallel, full-control operation

5.4.1 DIP switch setting

The DIP switch settings apply to CU 212.

- Switches **1, 2** and **3**: System type



This setting encodes the system type.
CU 212: Four float switches in parallel, full-control operation.

- Switch **4**: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

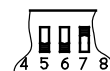
- Switches **5, 6** and **7**: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.
The pump must not run dry.

0 sec.



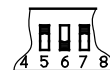
60 sec.



15 sec.



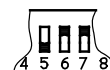
90 sec.



30 sec.



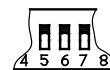
120 sec.



45 sec.



180 sec.



- Switch **8**:



Switch 8 has no function in connection with this system type, but keep the setting in this position.

- Switch **9**: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch **10**: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



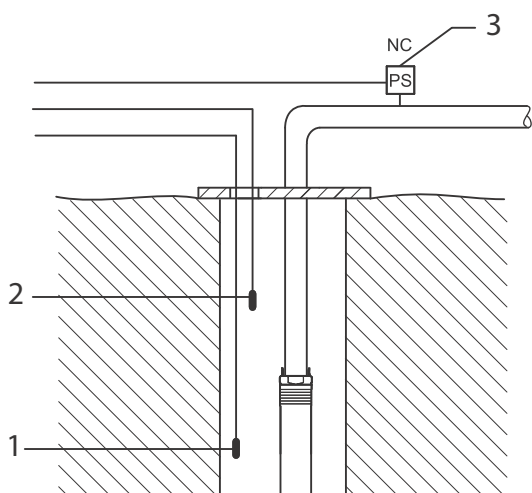
Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.5 Two electrodes

Use **CU 211** for systems with two electrodes. The pump is controlled by the liquid level in the borehole.

- The electrode, pos. 1, is a reference electrode.
- When the electrode, pos. 2, registers liquid, the settable start delay sets in, and after that the pump is started.
- When the electrode, pos. 2, no longer registers liquid, the pump is stopped.
- The pressure switch, pos. 3, will stop the pump if the pump discharge pressure exceeds the pressure switch cut-out pressure. The pump will restart automatically when the pressure falls to the pressure switch cut-in pressure if the electrode, pos. 2, also registers liquid.



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Fig. 13 System with one pump and two electrodes

5.5.1 DIP switch setting

The DIP switch settings apply to CU 211.

- Switches 1, 2 and 3: System type



This setting encodes the system type.
CU 211: Two electrodes.

- Switch 4: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

- Switches 5, 6, 7 and 8: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.

The pump must not run dry.

1 min.



30 min.



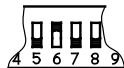
2 min.



35 min.



3 min.



40 min.



5 min.



45 min.



10 min.



50 min.



15 min.



55 min.



20 min.



60 min.



25 min.



65 min.



- Switch 9: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch 10: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



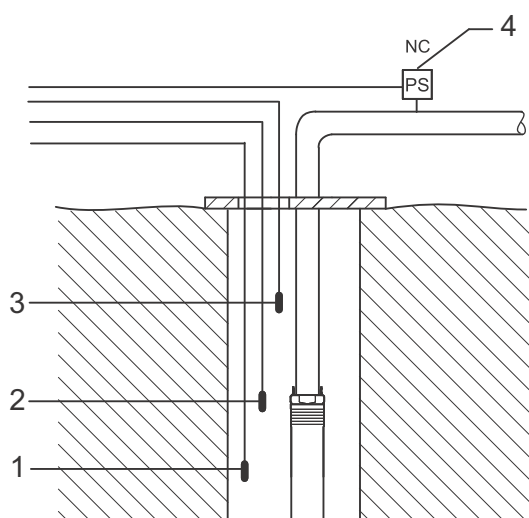
Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.6 Three electrodes

Use **CU 211** for systems with three electrodes. The pump is controlled by the liquid level in the borehole.

- The electrode, pos. 1, is a reference electrode.
- When the electrode, pos. 3, registers liquid, the pump is started.
- When the electrode, pos. 2, no longer registers liquid, the pump is stopped. It is possible to set a stop delay which delays the stop of the pump.
- The pressure switch, pos. 4, will stop the pump if the pump discharge pressure exceeds the pressure switch cut-out pressure. The pump will restart automatically when the pressure falls to the pressure switch cut-in pressure if the electrode, pos. 3, also registers liquid.



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Fig. 14 System with one pump and three electrodes

5.6.1 DIP switch setting

The DIP switch settings apply to CU 211.

- Switches **1, 2** and **3**: System type



This setting encodes the system type.
CU 211: Two electrodes.

- Switch **4**: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

- Switches **5, 6** and **7**: Stop delay

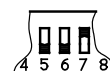
The **stop delay** is the time from the stop signal is given until the pump is stopped.

The pump must not run dry.

0 sec.



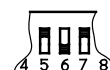
60 sec.



15 sec.



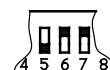
90 sec.



30 sec.



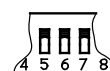
120 sec.



45 sec.



180 sec.



- Switch **8**:



Switch 8 has no function in connection with this system type, but keep the setting in this position.

- Switch **9**: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch **10**: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

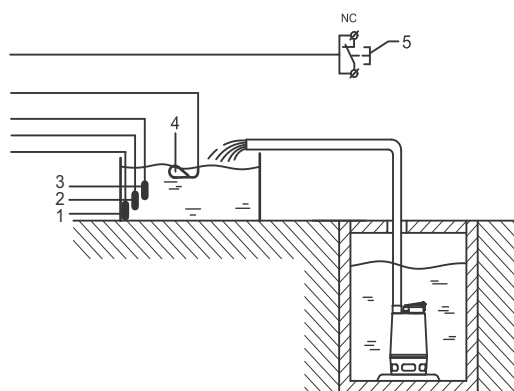
5.7 Three electrodes and one float switch

Use **CU 211** in systems with three electrodes and one float switch in applications such as tank filling. The pump is controlled by the liquid level in the tank.

- The electrode, pos. 1, is a reference electrode.
- When the electrode, pos. 2, no longer registers liquid, the pump is started.
- When the electrode, pos. 3, registers liquid, the pump is stopped. It is possible to set a stop delay which delays the stop of the pump.
- When the float switch, pos. 4, is lifted up by the liquid, the high-level alarm is activated.
- The pump can be stopped by means of the manual on/off switch, pos. 5.

Note

The three electrodes may be replaced by float switches.



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Fig. 15 One pump, three electrodes and one float switch in a tank filling system

5.7.1 DIP switch setting

The DIP switch settings apply to CU 211.

- Switches **1, 2** and **3**: System type



This setting encodes the system type.
CU 211: Three electrodes and one float switch in a tank filling system.

- Switch **4**: Start delay and automatic test run



This setting activates a start delay in the range from 0 to 255 sec. (random) after the power supply has been switched on.
Automatic test run is carried out every 24 hours.



This setting causes the pump to start immediately after the mains supply has been switched on.
No automatic test run.

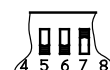
- Switches **5, 6** and **7**: Stop delay

The **stop delay** is the time from the stop signal is given until the pump is stopped.
The pump must not run dry.

0 sec.



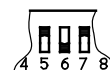
60 sec.



15 sec.



90 sec.



30 sec.



120 sec.



45 sec.



180 sec.



- Switch **8**:



Switch 8 has no function in connection with this system type, but keep the setting in this position.

- Switch **9**: Alarm resetting



This setting ensures that alarm signals from external alarm devices and the built-in buzzer are reset automatically. However, an alarm signal will only be reset if the cause of the fault no longer exists.



At this setting, the alarm signal must be reset manually by means of the reset button.
See section 2.1.1 *Resetting of alarm*.

- Switch **10**: Restarting



This setting enables automatic restarting after the motor PTC resistor/thermal switch has caused the pump to cut out. However, restarting will not happen until the motor has cooled off again.



Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.



This setting requires the pumps to be restarted manually after the motor PTC resistor/thermal switch has caused the pump to be cut out. Reset by means of the ON-OFF-AUTO selector switch.



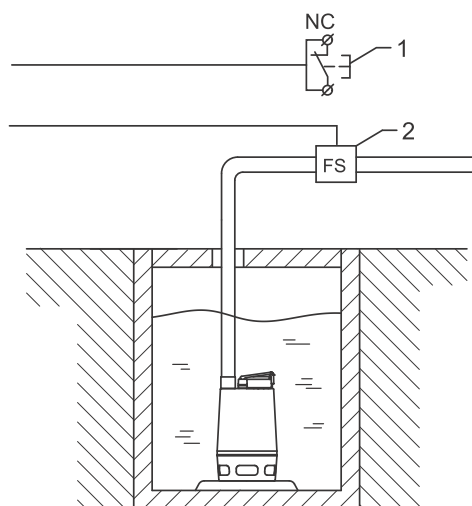
Warning

When the pump(s) connected is (are) used in potentially explosive environments, switch 10 must not be set to this position.

5.8 Flow switch

Use **CU 211** in systems controlled by a flow switch in applications such as draining systems. The pump is controlled by the flow in the discharge line.

- The pump is stopped when the flow switch, pos. 2, registers that there is no liquid flow.
- When the pump has stopped, a settable start delay sets in, and after that the pump will try to restart. The restart attempt will be interrupted when the settable dry running time has elapsed if the flow switch still does not register flow.
- The pump can be restarted manually by means of the switch, pos. 1.



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Fig. 16 Flow switch in a draining system

5.8.1 DIP switch setting

The DIP switch settings apply to CU 211.

- Switches **1, 2** and **3**: System type

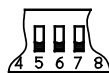


This setting encodes the system type.
CU 211: Flow switch in a draining system.

- Switches **4, 5** and **6**: Dry running time

The **dry running time** is the time the pump is allowed to run after start even if the flow switch registers no flow.
The pump is stopped again if the flow switch still does not register flow.

10 sec.



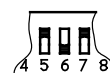
2 min.



20 sec.



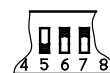
3 min.



40 sec.



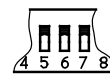
4 min.



1 min.



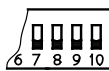
5 min.



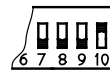
- Switches **7, 8, 9** and **10**: Start delay

The **start delay** is the time from the pump stopped until the pump attempts to restart.

No restart *



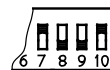
15 min.



1 min.



17 min.



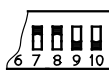
2 min.



20 min.



3 min.



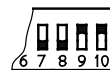
25 min.



5 min.



30 min.



7 min.



50 min.



10 min.



70 min.



12 min.



90 min.



* If the DIP switch was set to "no restart", the pump can only be restarted by means of the switch.

6. Installation

Before installation, check these points:

- does the variant correspond to the one ordered?
- does CU 211/CU 212 match the supply voltage and frequency of the installation site?
- has CU 211/CU 212 been damaged during the transport?



Warning

CU 211 and CU 212 must not be used for other purposes than the ones specified.



Warning

Before installing CU 211 and CU 212, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

The installation must be carried out by authorized personnel in accordance with local regulations.

All safety regulations must be observed on the installation site.

6.1 Location

Enclosure

CU 211 and CU 212 have enclosure class IP20. In order to reduce the pollution level to 2, CU 211 and CU 212 must be installed in a protective enclosure with minimum IPX4 enclosure class according to IEC 60529. The enclosure must be made of fire retardant material.

Ambient temperature

Ambient temperature: -30°C to $+50^{\circ}\text{C}$.

CU 211 and CU 212 must not be exposed to direct sunlight.

Installation altitude

CU 211 and CU 212 must be installed max. 2000 m above sea level.

Warning

Do not install CU 211 and CU 212 in potentially explosive environments.

If the controlled pumps are in potentially explosive environments, CU 211 and CU 212 must be used in combination with float switches.

Float switches located in potentially explosive environments must be connected via an EEx barrier, such as Grundfos no. 96440300. The EEx barrier itself must not be installed in potentially explosive environments.

Equipment/material located in potentially explosive environments must be approved for this in each individual situation.

Likewise the cabling into potentially explosive environments must be made according to current regulations.



Note

Use EEx approved float switches, such as Grundfos no. 96003332 or 96003695.

Mounting and removal

CU 211 and CU 212 are designed for mounting on a 35 mm DIN rail (EN 50 022).

Mount CU 211 and CU 212 in this way:

1. Hold the bottom of the control unit up against the DIN rail.
2. Push the unit upwards.
3. Tip the unit into place. See fig. 17.

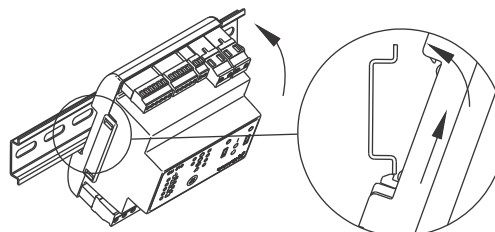


Fig. 17 Mounting of CU 211 and CU 212

Remove CU 211 and CU 212 in this way:

1. Push the unit upwards.
2. Tip the unit free of the DIN rail.

7. Electrical connection

Supply voltage: Single-phase = 230 V $\pm 10\%$ 50/60 Hz
Three-phase = 400 V $\pm 10\%$ 50/60 Hz.

Supply current: Maks. 250 mA. See also section 8. *Overview of terminals.*

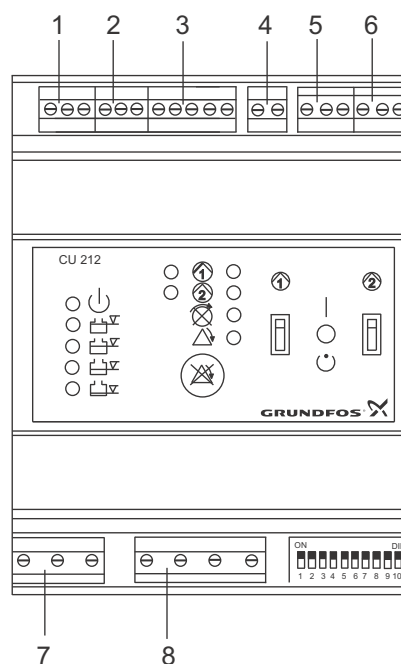


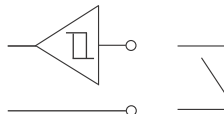
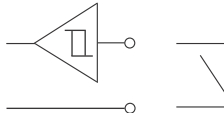
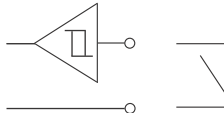
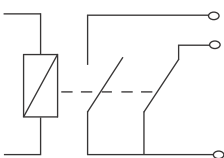
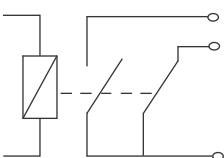
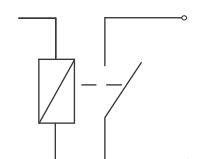
Fig. 18 CU 212 with overview of terminals

TM04 3069 3608

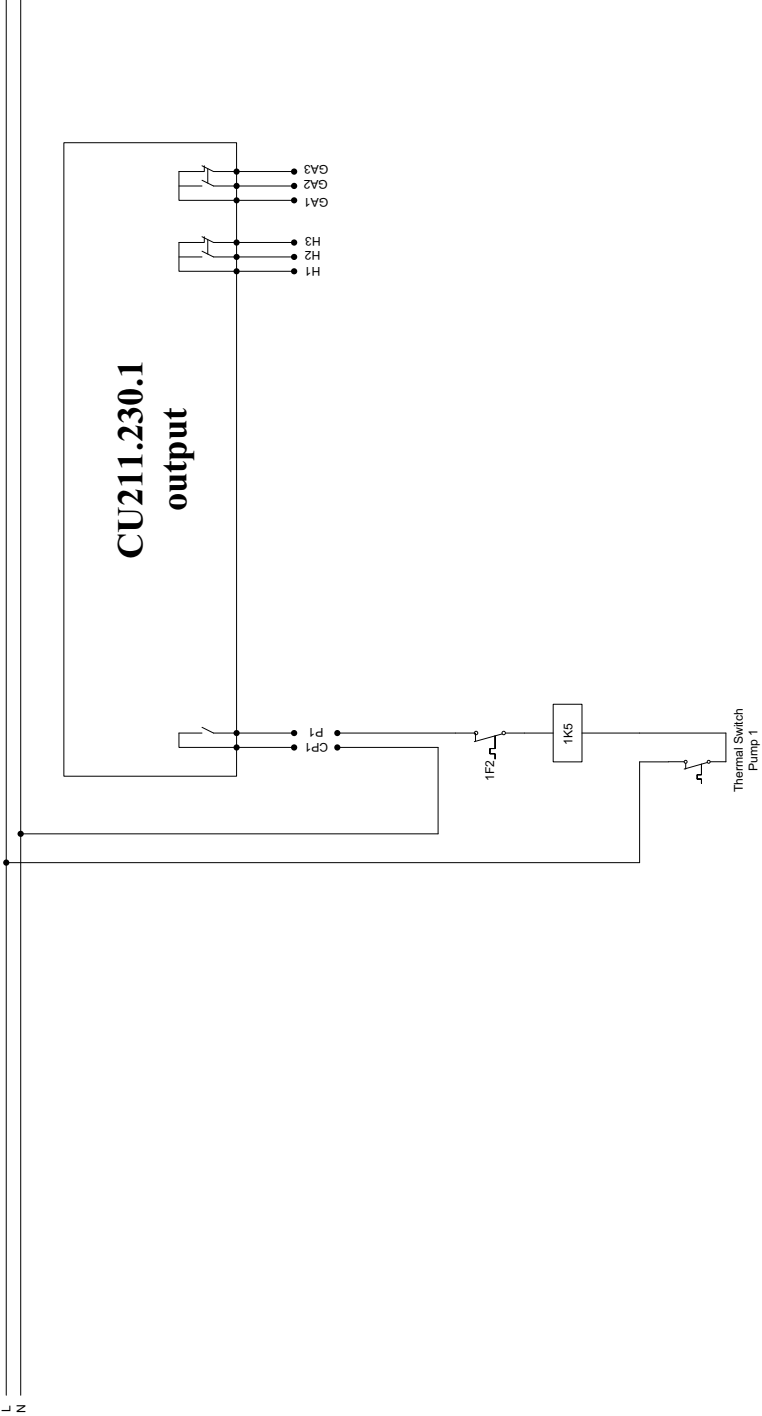
TM04 3133 3808

8. Overview of terminals

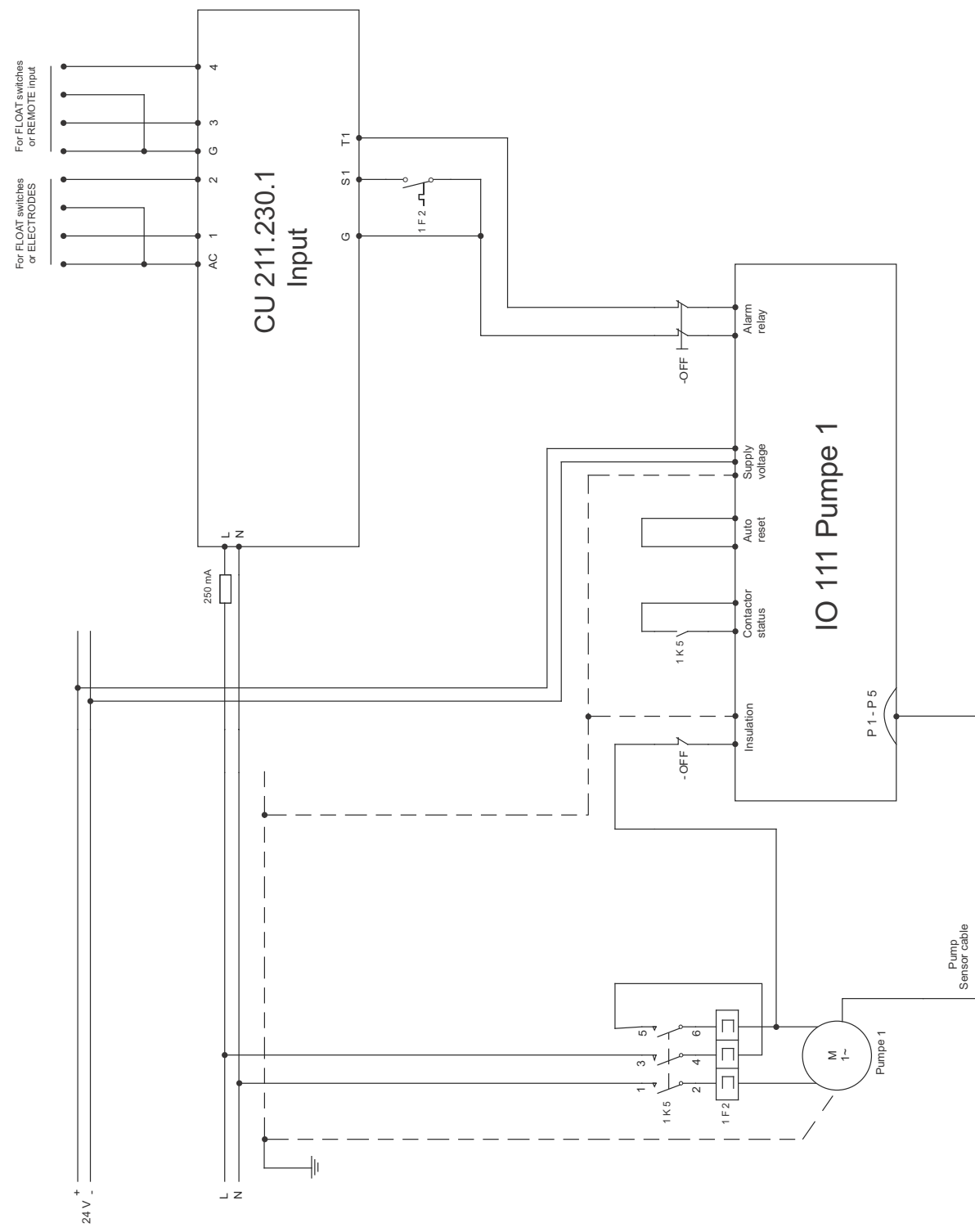
The designations of the terminals refer to the wiring diagrams on the following pages.

Pos.	Terminal	Description	Electrical data	Function	Diagram
1	AC	Supply output for level 1 and 2	13-18 AC / 12 VDC	The level 1 and 2 inputs are designed for either AC or DC transmitters, such as electrodes or float switches.	
1	1	Terminal for level 1			
1	2	Terminal for level 2	1 mA		
2	G	Earth (GND) for level 3 and 4	0 V	The level 3 and 4 inputs are designed for DC transmitters, such as float switches. A max. cable length of 100 metres is recommended.	
2	3	Terminal for level 3			
2	4	Terminal for level 4	5 V output max. 0.5 mA		
3	G	Frame connection for S1-2 and T1-2	0 V	S1 is input for alarm from overload relay. T1 is input from a thermal switch fitted inside the pump. S2 only applies to two-pump systems; otherwise the same as S1. T2 only applies to two-pump systems; otherwise the same as T1.	
3	S1	Terminal for motor protector	5 V output max. 0.5 mA		
3	T1	Terminal for thermal circuit breaker	5 V output max. 2.2 mA		
3	S2	Terminal for motor protector	5 V output max. 0.5 mA		
3	T2	Terminal for thermal circuit breaker	5 V uoutput max. 2,2 mA		
5	H3	Teminal for NC-contact for high-level alarm	Max 230 VAC 2 A	Depending on the application and the DIP switch setting, one of the level inputs can trip the high-level alarm.	
5	H2	Frame connection for high-level alarm			
5	H1	Terminal for NO-contact for high-level alarm			
6	GA1	Terminal for NC-contact for common alarm	Max. 230 VAC 2 A	All alarms trip the alarm relay.	
6	GA2	Frame connection for common alarm			
6	GA3	Terminal for NO-contact for common alarm			
7	L1	Supply input. Only three-phase version	400 V	Mains supply input. In three-phase versions the rated voltage is 400 V on L1, L2 and L3. In single-phase versions the rated voltage is 230 V on L and N.	
7	L2/L	Supply input	400 V/230 V		
7	L3/N	Supply input	400 V/230 V		
8	CP1	Terminal for start/stop of pump	Max. 400 V 2 A	P1 is an NO-contact designed for start and stop of pump 1. Depending on the DIP switch setting, the relay will close according to the signals from the level inputs.	
8	P1	Terminal for start/stop of pump 1			
8	CP2	Terminal for start/stop of pump	Max. 400 V 2 A	P2 only applies to two-pump systems; otherwise the same as P1.	
8	P2	Terminal for start/stop of pump 2			

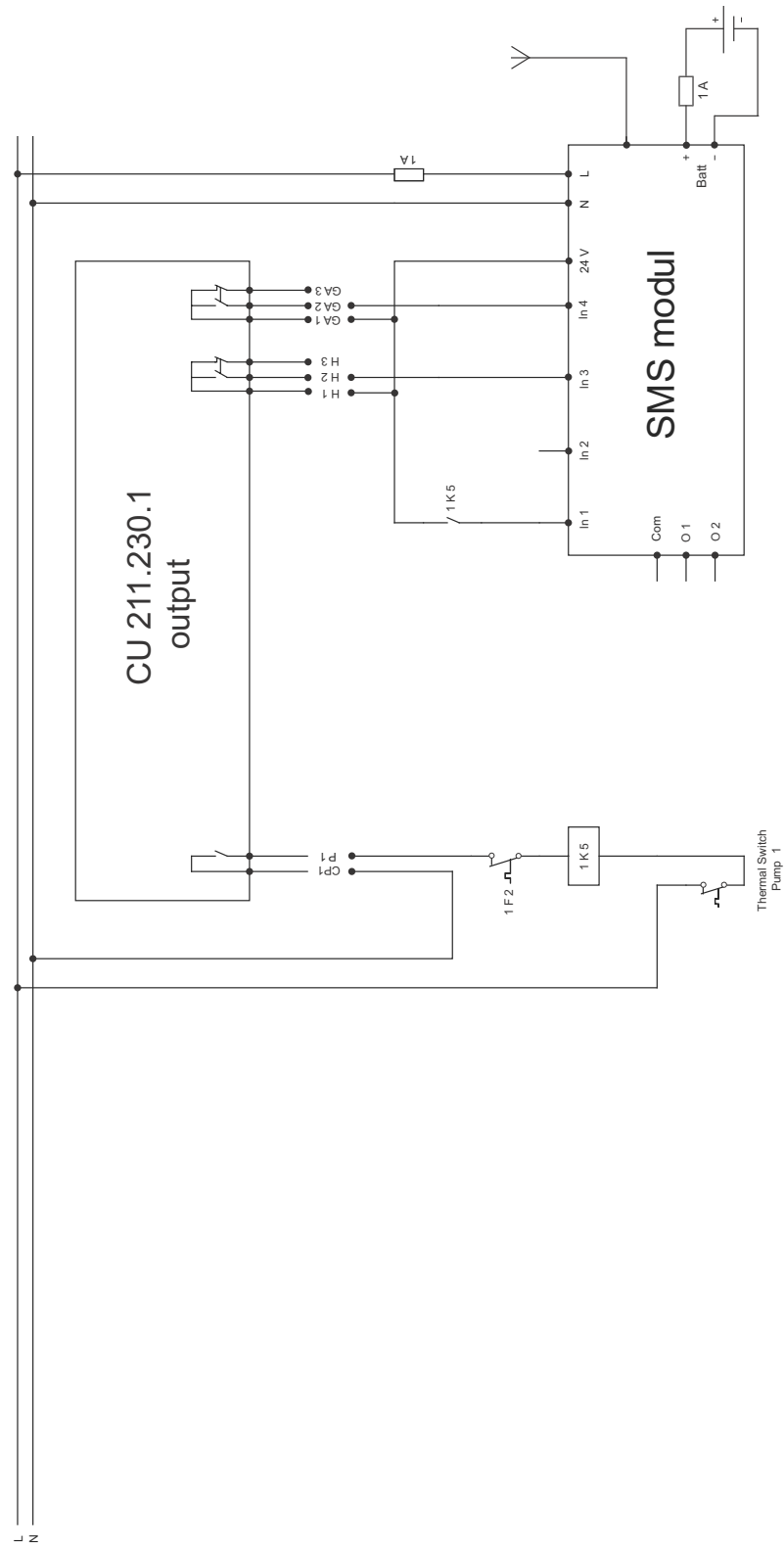




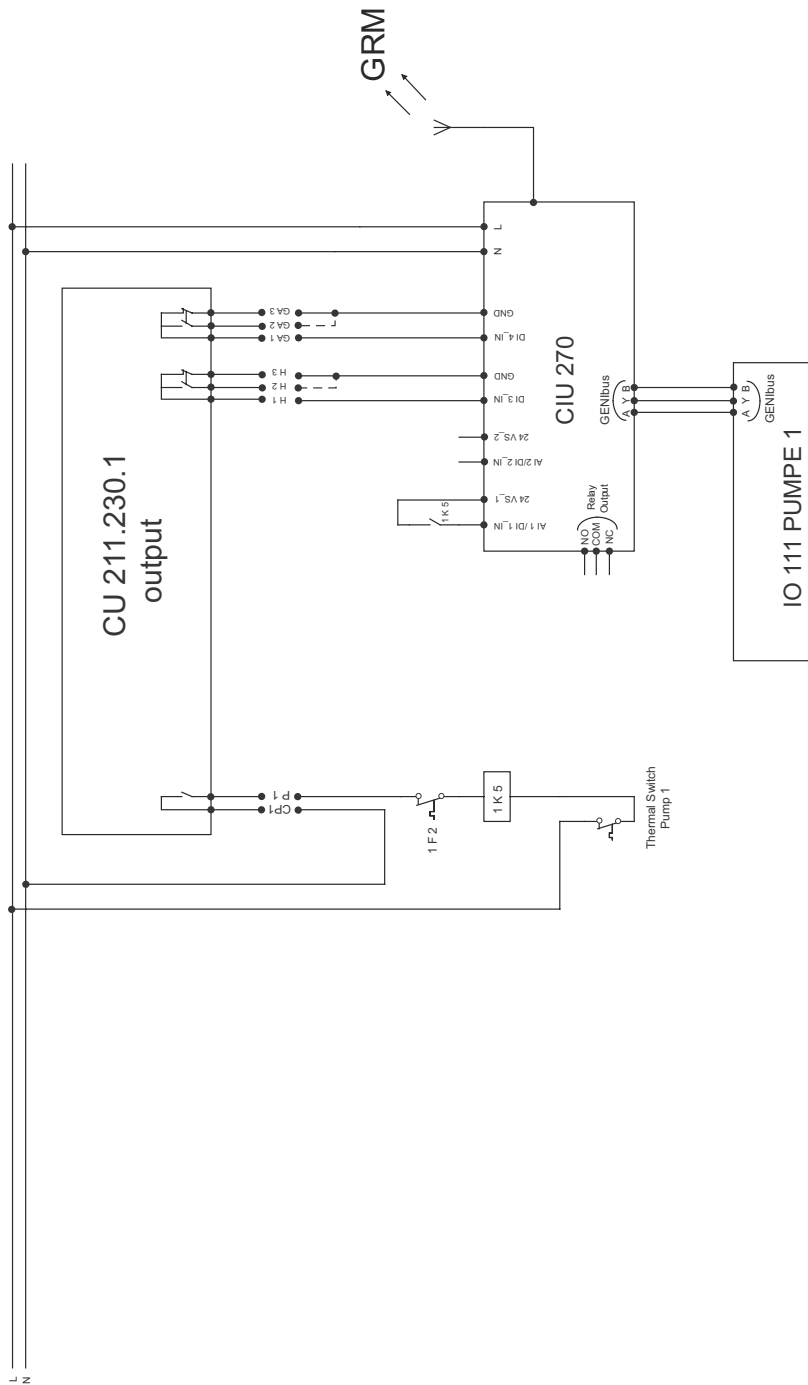
9.3 CU 211, single-phase, inputs from IO 111



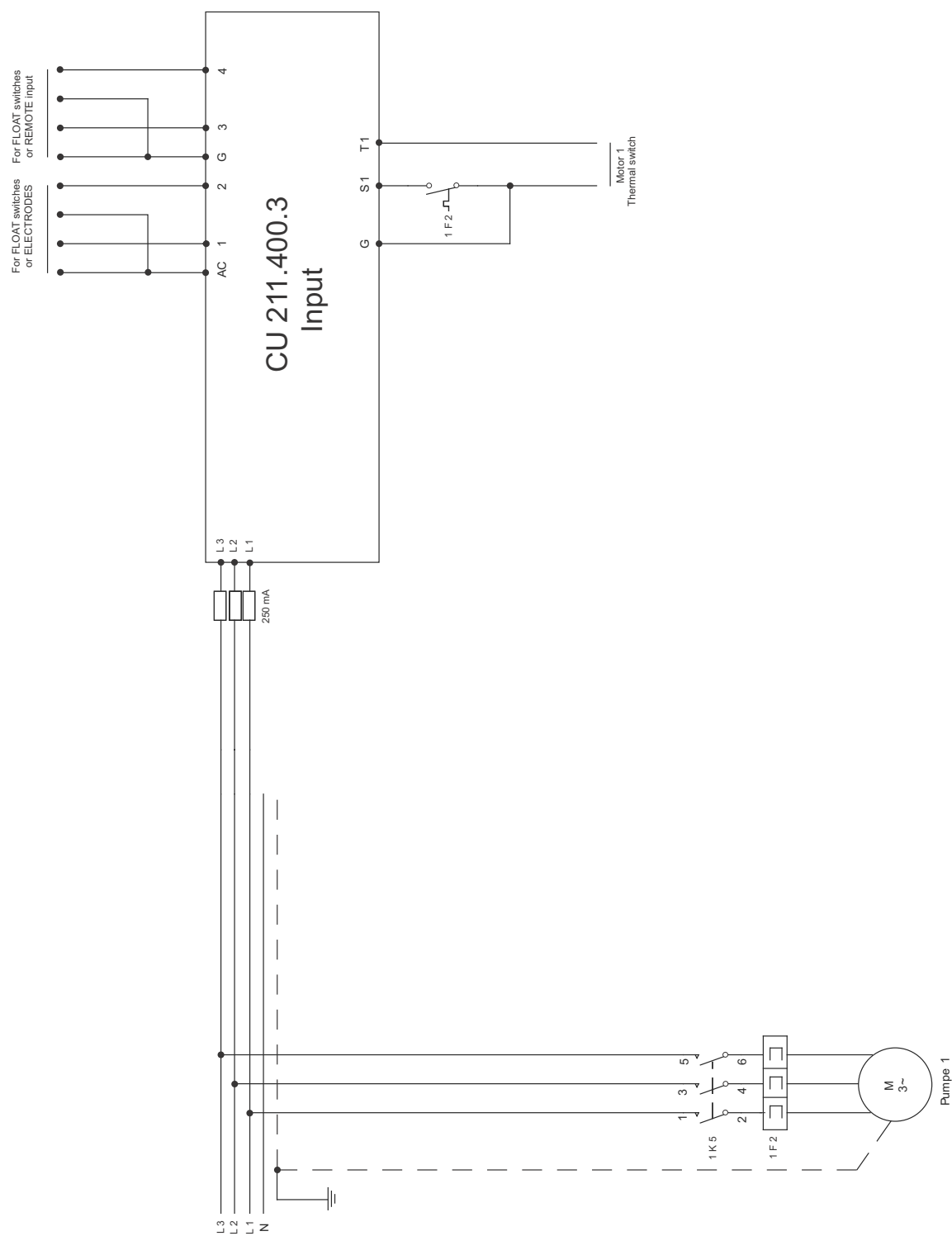
9.4 CU 211, single-phase, outputs for modem



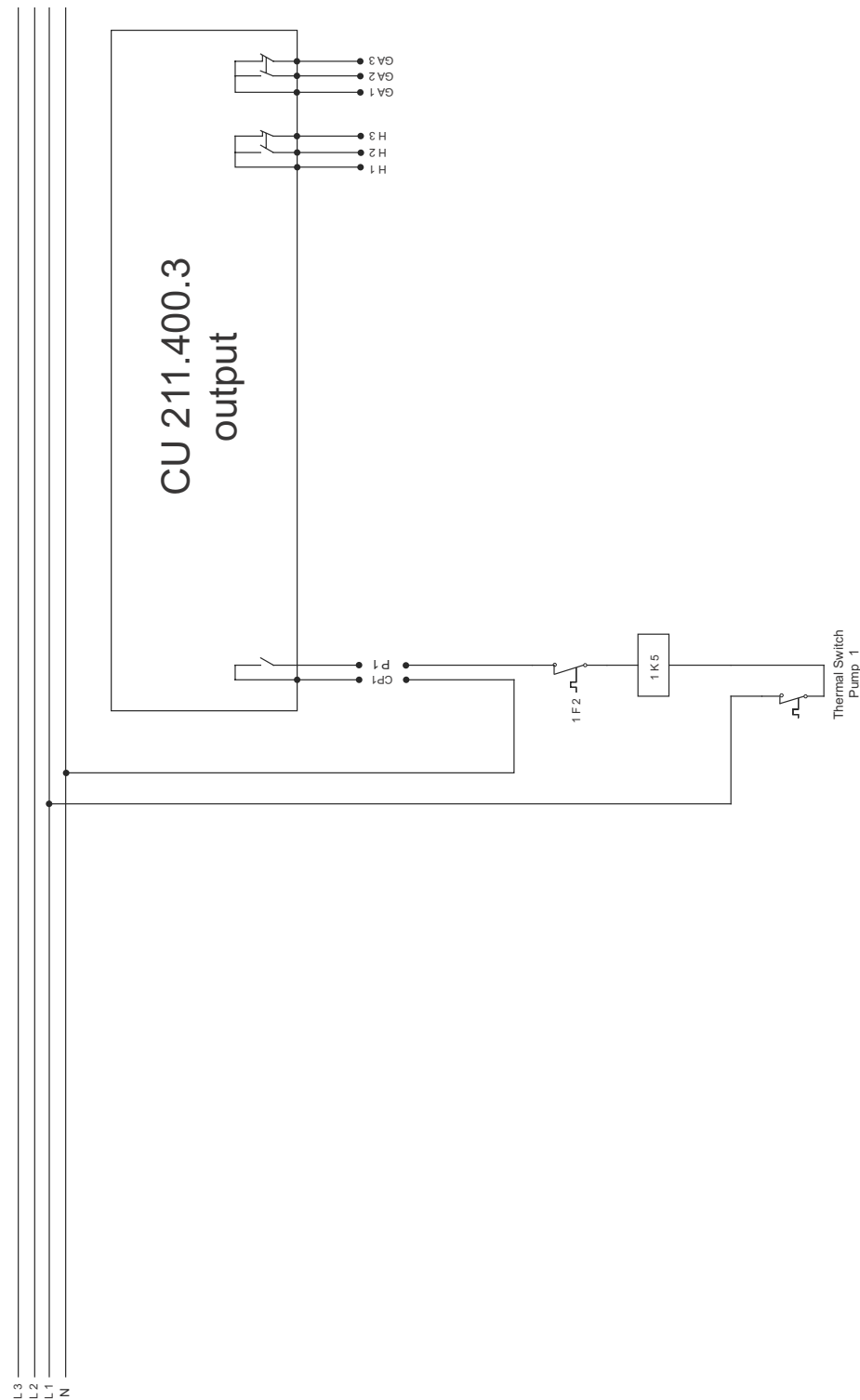
9.5 CU 211, single-phase, outputs for CIU 270



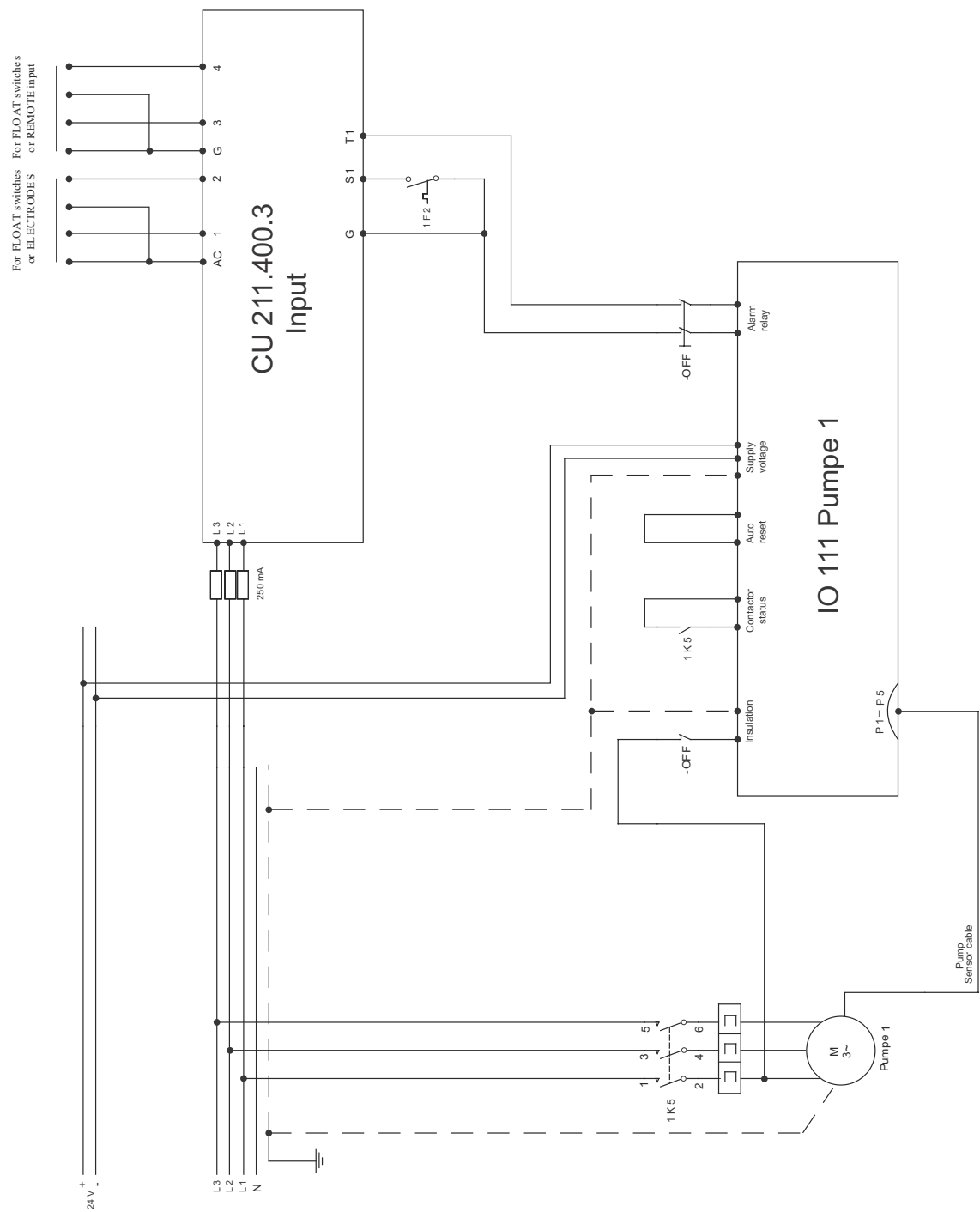
9.6 CU 211, three-phase, inputs



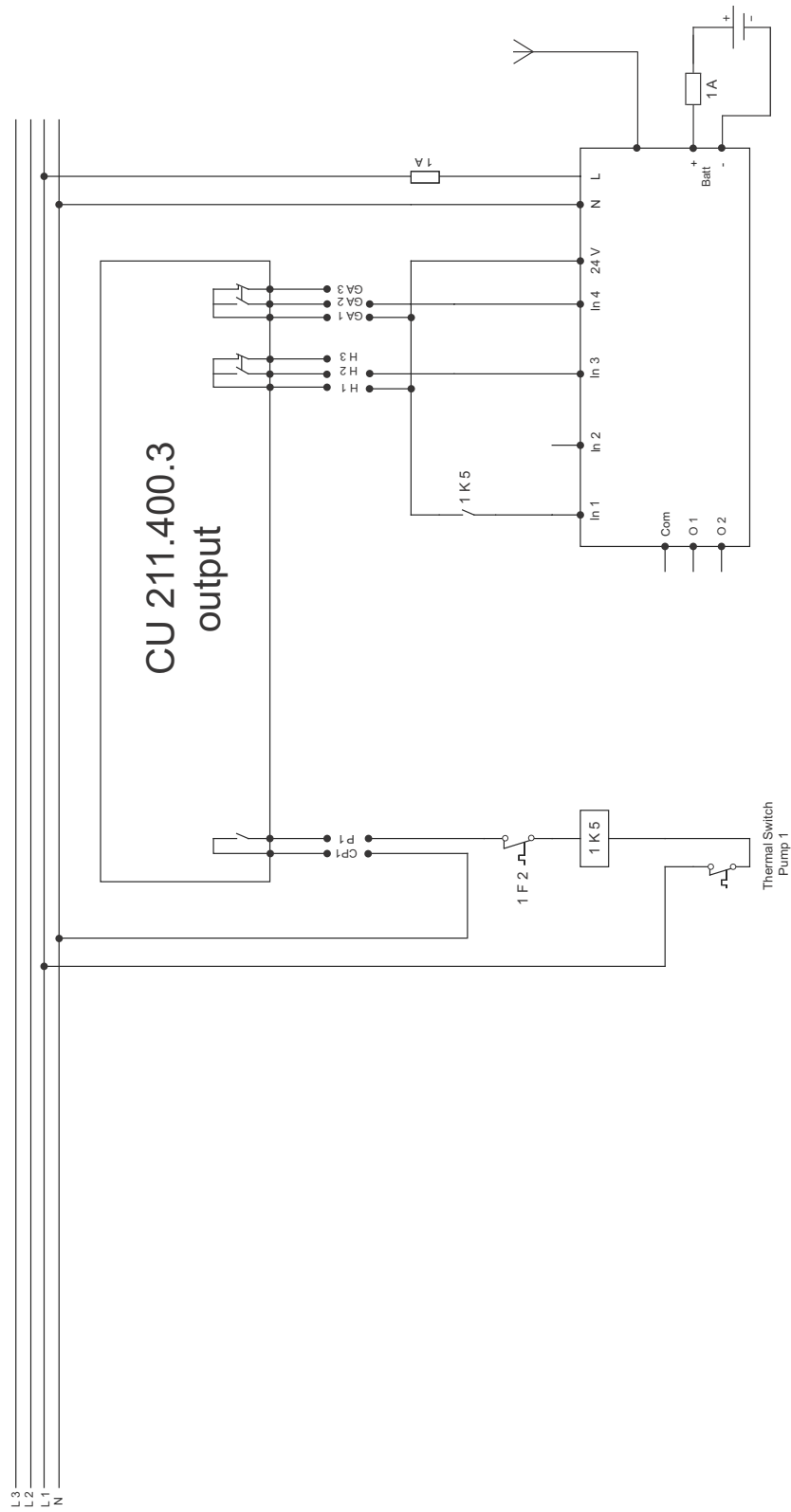
9.7 CU 211, three-phase, outputs



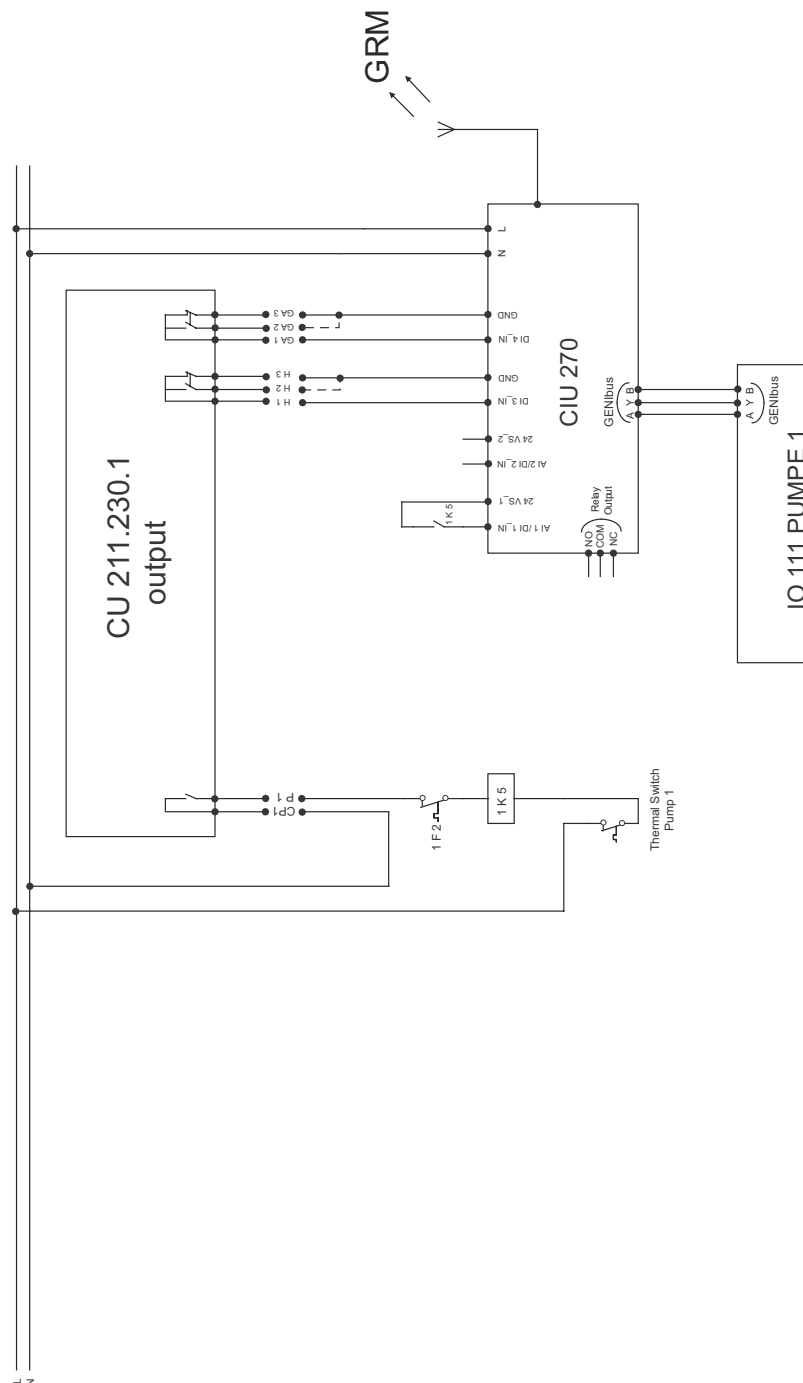
9.8 CU 211, three-phase, inputs from IO 111



9.9 CU 211, three-phase, outputs for modem

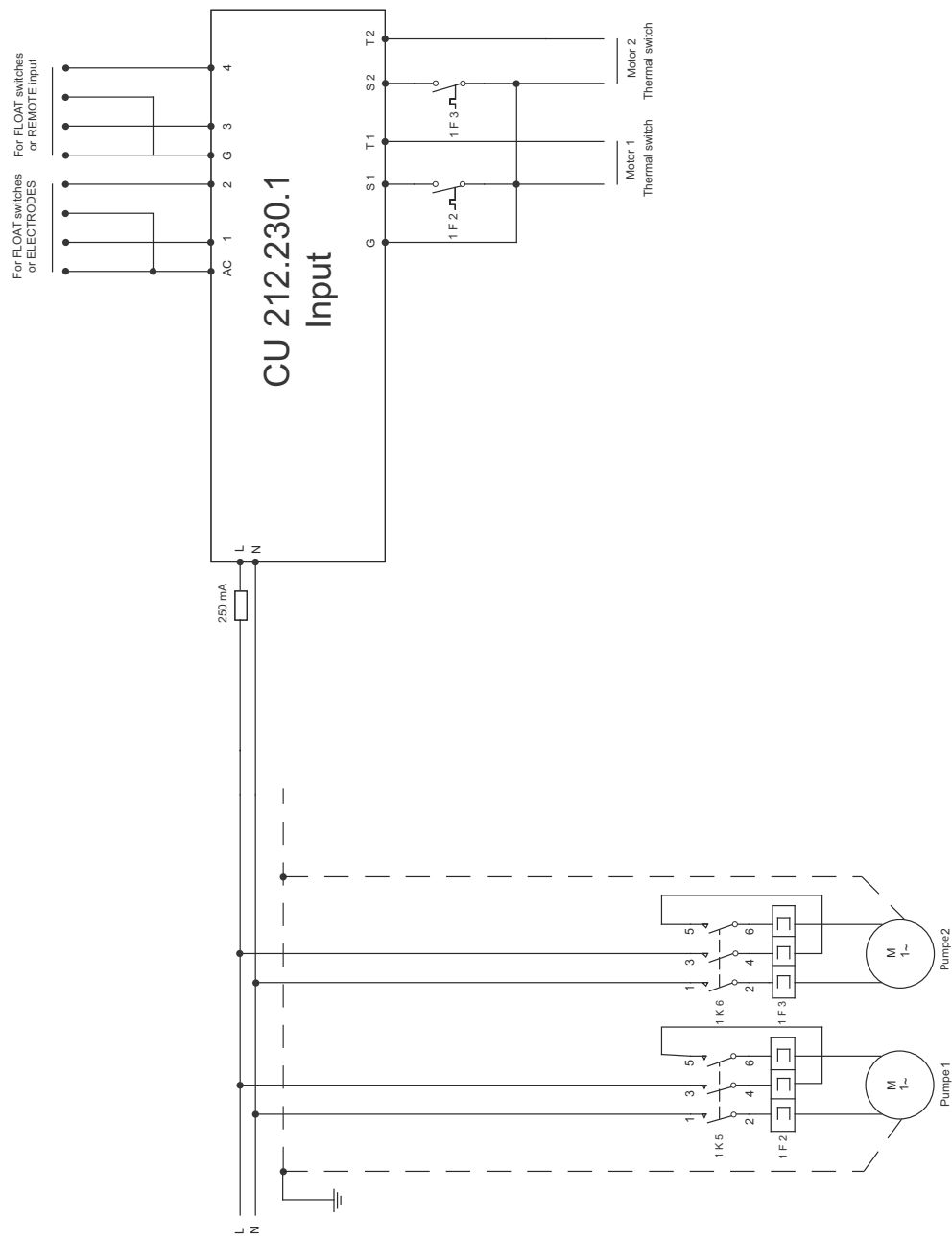


9.10 CU 211, three-phase, outputs for CIU 270

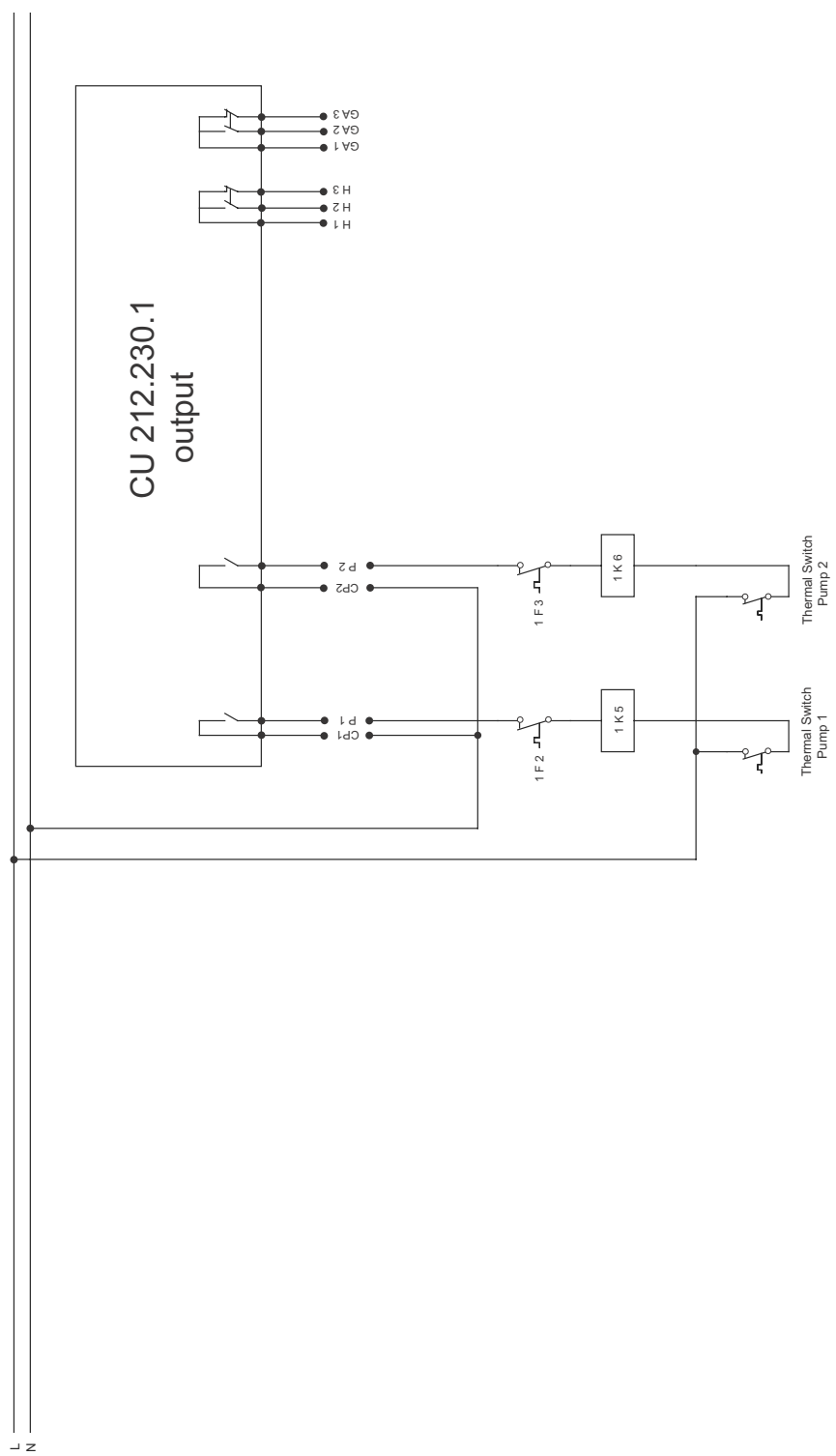


10. Wiring diagrams CU 212

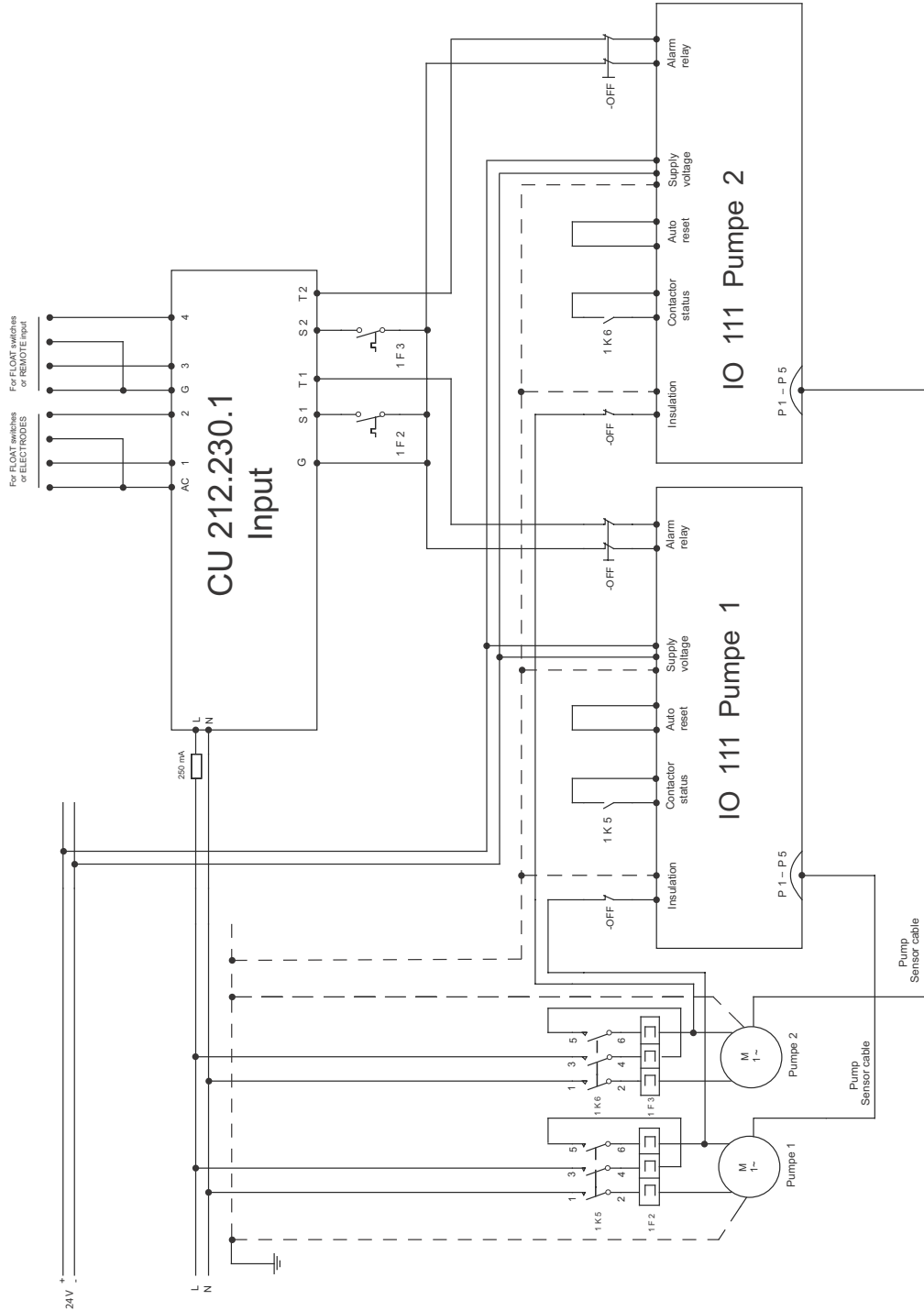
10.1 CU 212, single-phase, inputs

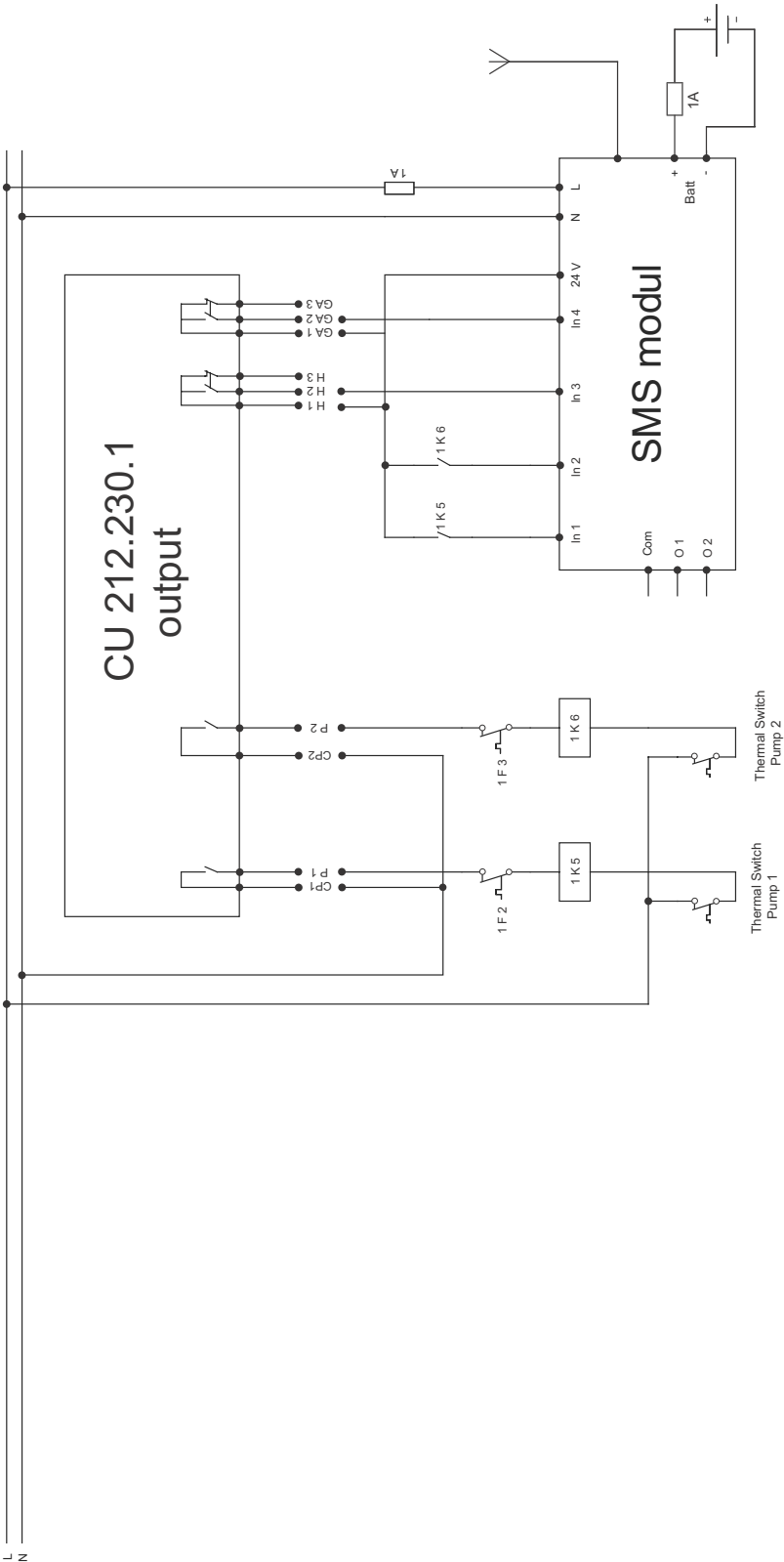


10.2 CU 212, single-phase, outputs

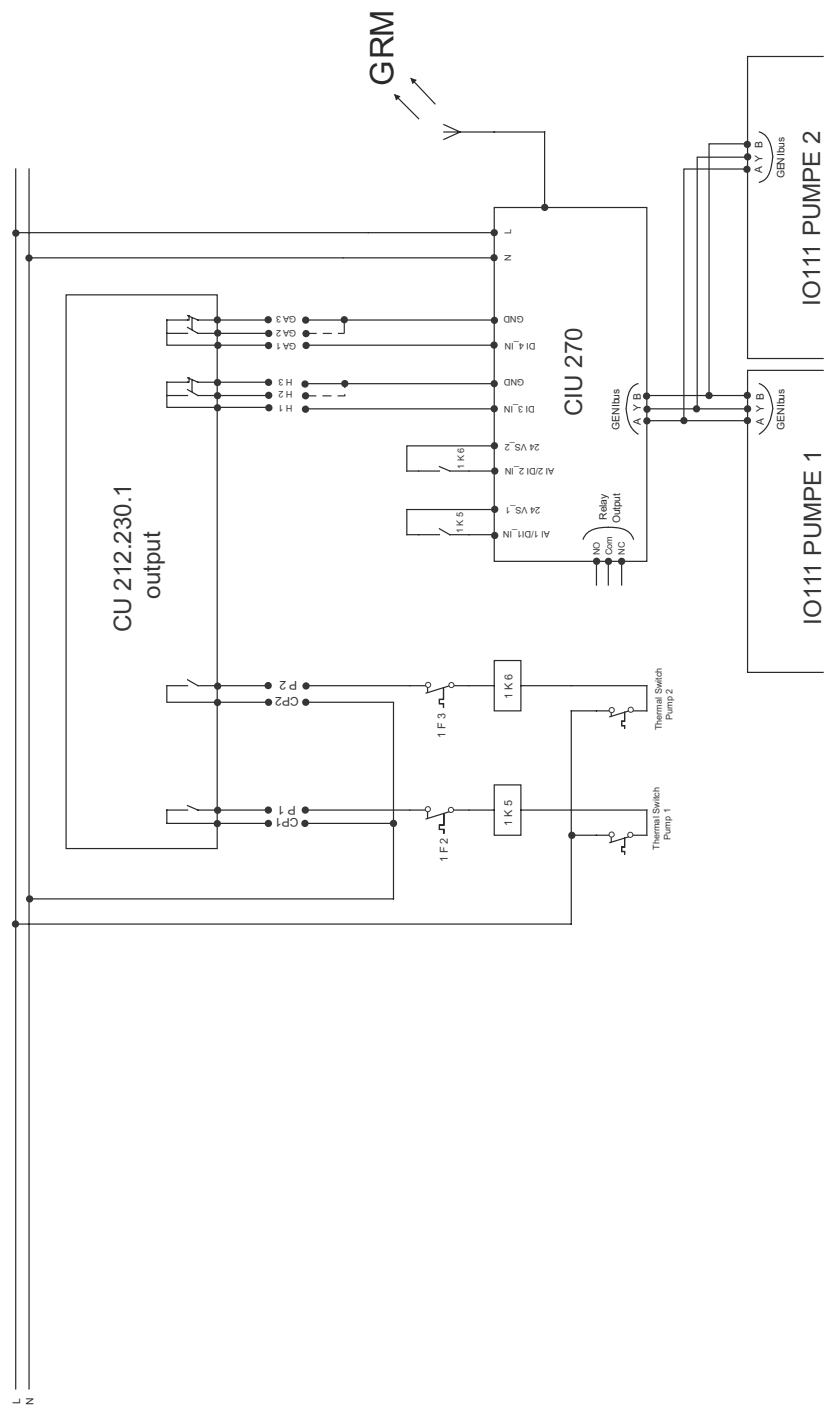


10.3 CU 212, single-phase, inputs from IO 111

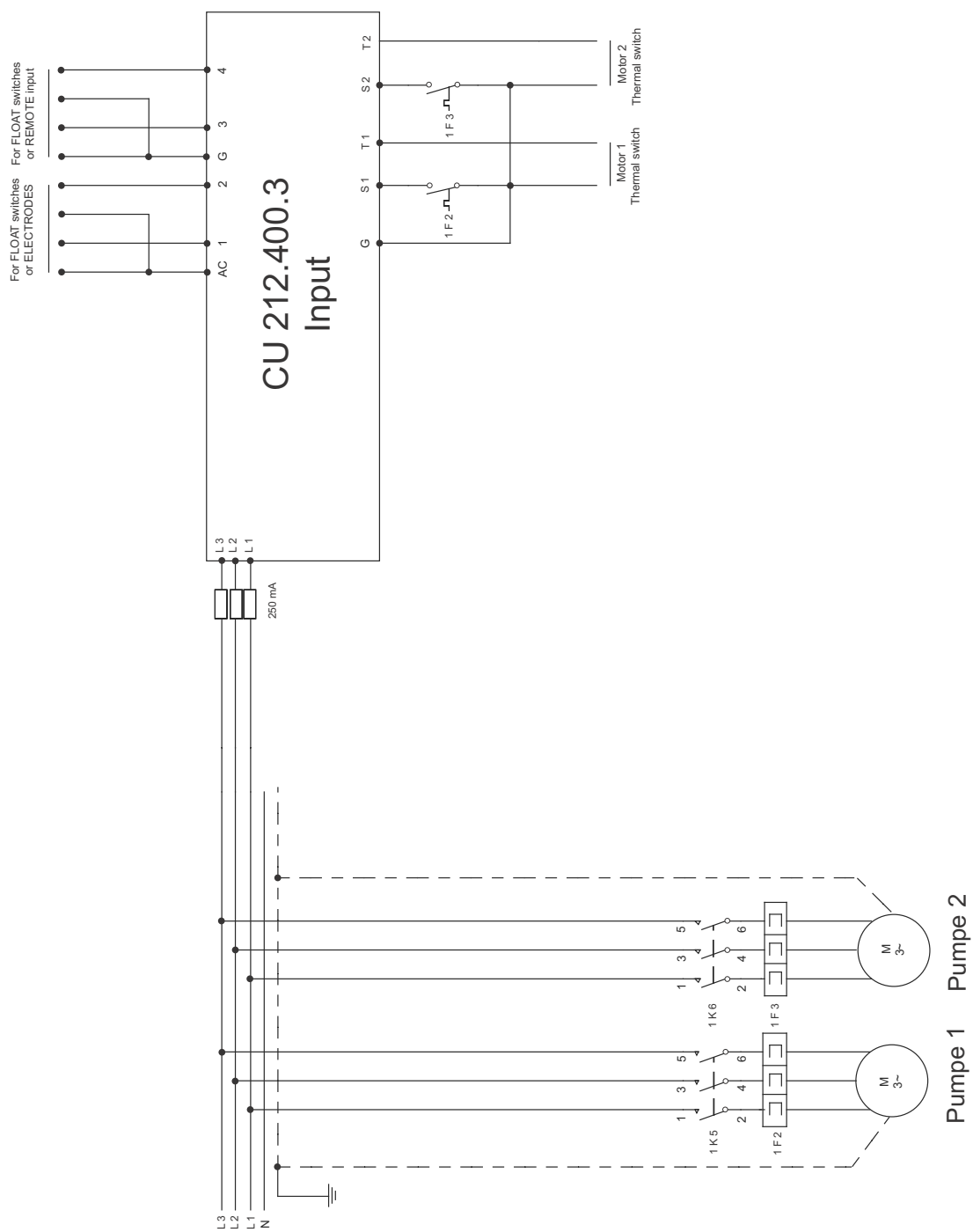




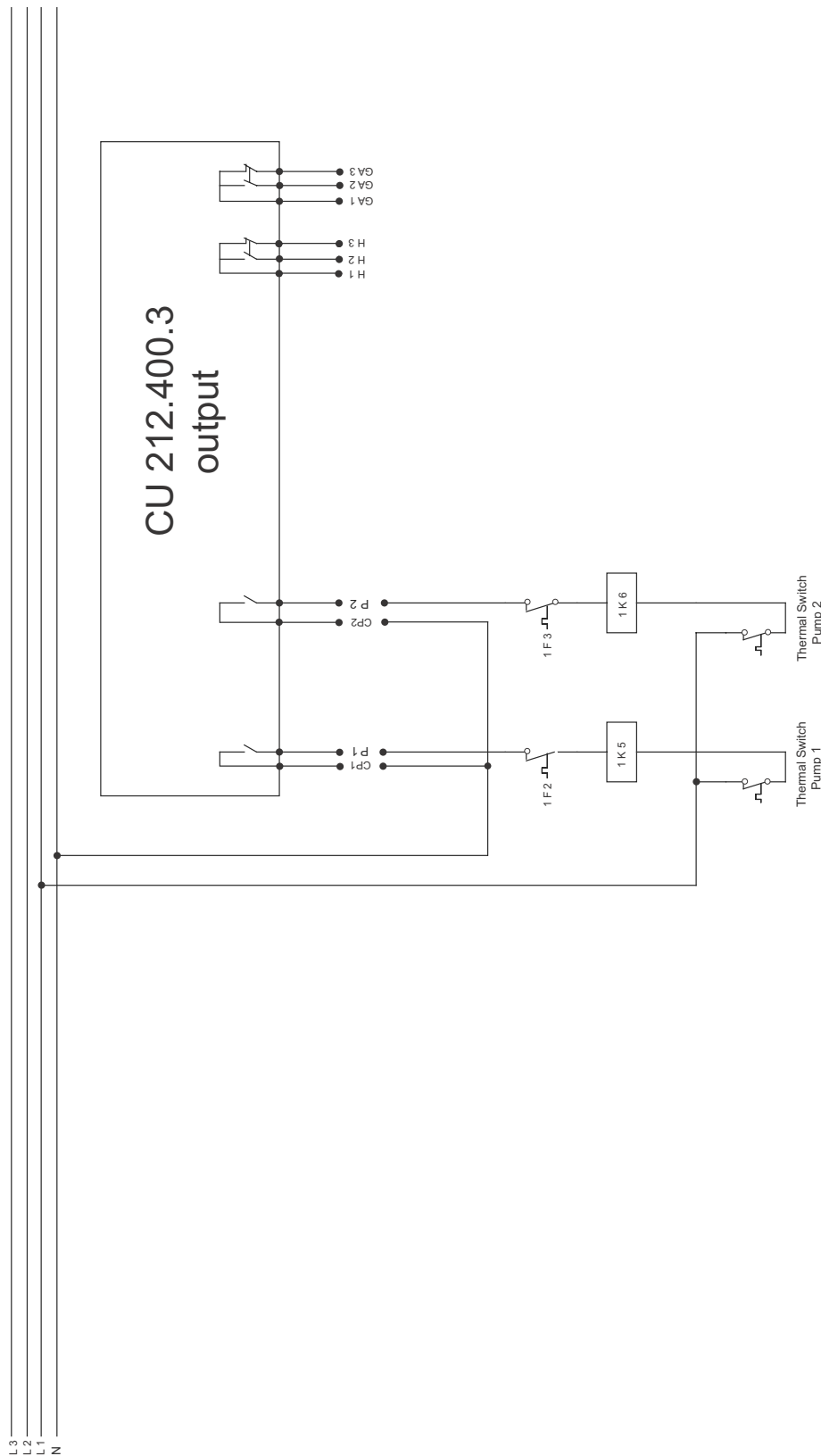
10.5 CU 212, single-phase, outputs for CIU 270



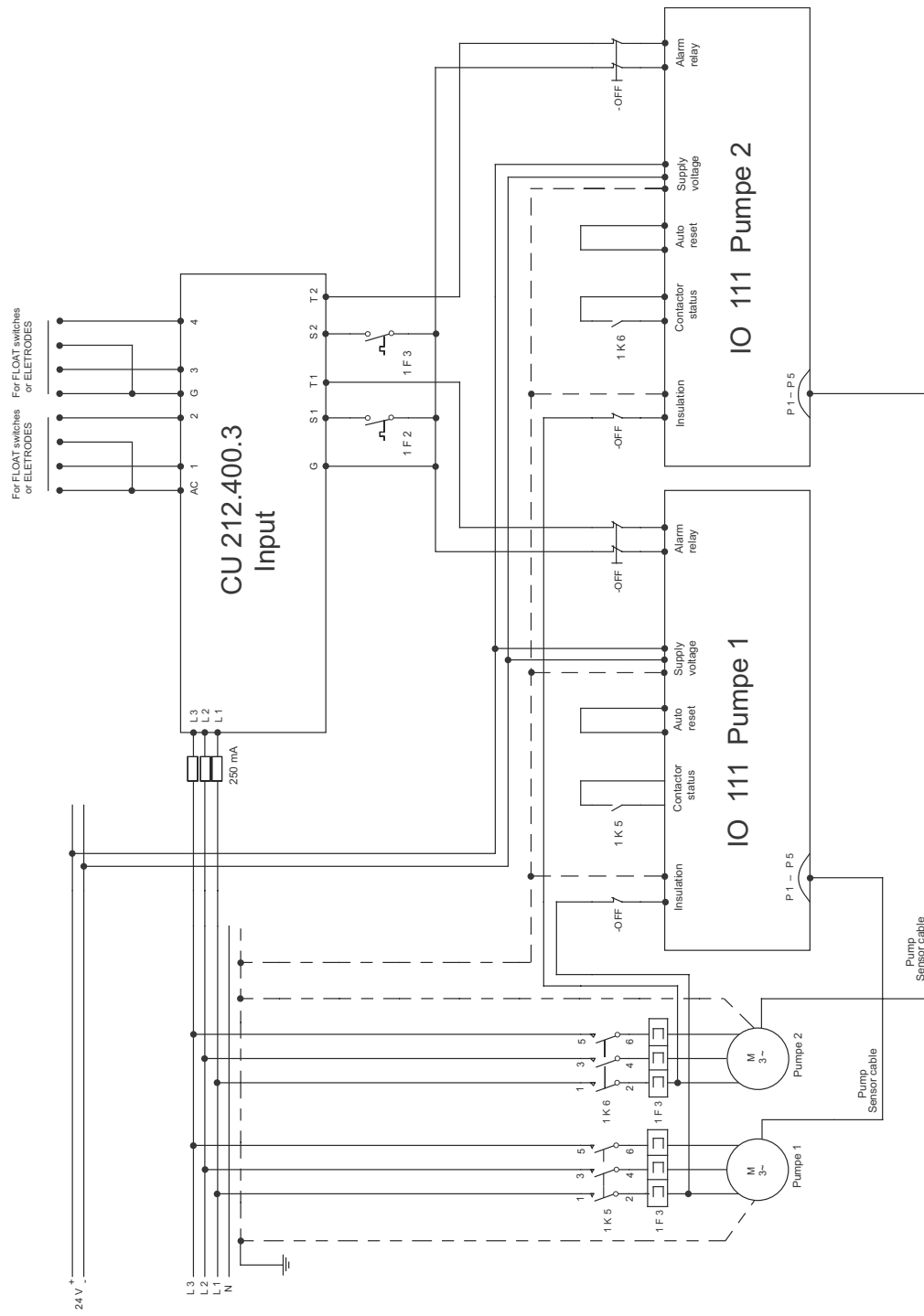
10.6 CU 212, three-phase, inputs



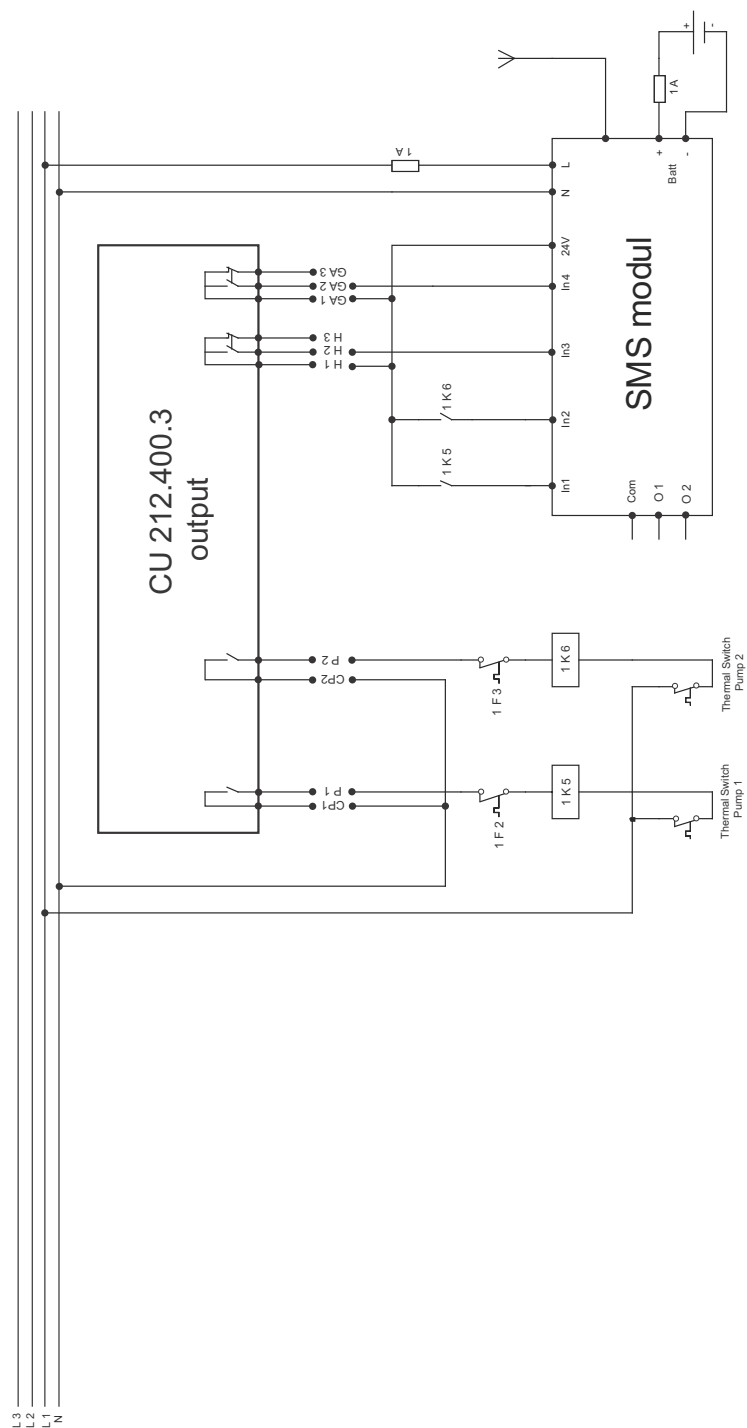
10.7 CU 212, three-phase, outputs



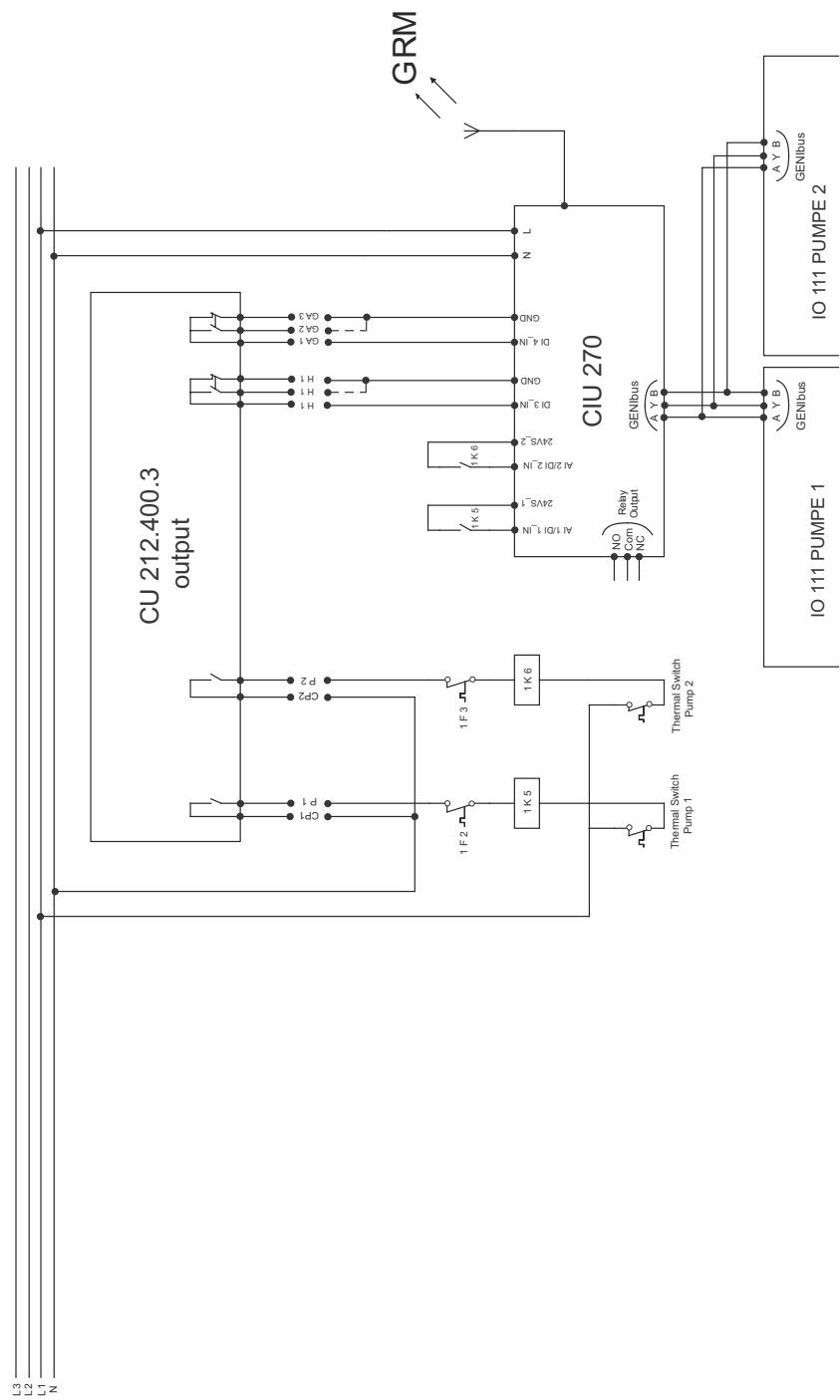
10.8 CU 212, three-phase, inputs from IO 111



10.9 CU 212, three-phase, outputs for modem



10.10 CU 212, three-phase, outputs for CIU 270



11. Maintenance

CU 211 and CU 212 require no maintenance. CU 211 and CU 212 may only be cleaned with a dustless cloth.

12. Service

CU 211 and CU 212 cannot be serviced. In case of faults in the control unit, replace it.

13. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

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